

MAE 242: Robot Motion Planning

Homework #4

Homework Instructions:

This is homework 4 analytical. Please recall the collaboration policy, and include in your solution the names of the other students you may have collaborated with to do the homework. Recall that you have to turn in your own solutions.

Homework exercises:

1. (LQR) Consider the following discrete-time LTI system:

$$x_{t+1} = Ax_t + Bu_t, \quad y_t = Cx_t,$$

where y_t is the system measurement (observation), and

$$C = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^\top, A = \begin{bmatrix} 1 & 0.4 & 0 & 0 \\ -0.6 & 1 & 0.4 & 0 \\ 0 & 0.4 & 1 & -0.6 \\ 0 & 0 & 0.4 & 1 \end{bmatrix}.$$

Let $Q = C^\top C$ and $R = 1$ and define a stage cost $g(x_t, u_t) = x_t^\top Q x_t + u_t^\top R u_t$.

Now, for any t , suppose that we solve an LQR problem to minimize the T -horizon cost $J(x_t) = \sum_{k=t}^{t+T-1} g(x_k, u_k)$. Observe that, since our system is linear and time-invariant, the optimal linear policy $u_t = K_T x_t$, has a gain, K_T , independent of t . Therefore, by applying $u_t = K_T x_t$ for each t , we are applying a policy that behaves optimally wrt T steps into the future.

Answer the following:

- (a) “As we increase T , the receding horizon control becomes less myopic and greedy”. Do you agree? Please explain carefully and intuitively in words.
- (b) “If the horizon is too short, the actions taken can result in an unstable closed-loop system”. Do you agree? Please explain carefully and intuitively in words.
- (c) What is the necessary and sufficient condition for the stability of the discrete-time closed-loop system $x_{t+1} = (A + BK_T)x_t$ (*Hint*: “spectral radius”)?
- (d) Calculate K_T for $T = 1, 2, \dots, 15$ and check which of them satisfy the conditions in the previous item. All you need to do is to write a simple code to implement the recursive formula that we derived for P_i and K_i for $i = 1, \dots, T$ for $T = 1, \dots, 15$ (set $P_1 = Q$). Then, check the condition in the previous item for each K_T . Plot the spectral radius of $A + BK_T$ versus T .
- (e) Look at your plot of the previous part. Is there a critical value of T , T_s , for which you observe stability of lack of stability afterwards?
- (f) Repeat the previous two parts for $Q' = \rho Q$, for $\rho = 1, 0.9, 0.8, \dots, 0.1$ and compute T_s for each ρ (let's call it $T_s(\rho)$). Plot $T_s(\rho)$ versus ρ . What do you observe? Explain carefully and intuitively.

2. As in the discrete planning part of the course, consider a state-transition graph, $G_T = (S, E)$, where actions at a state s are given by choosing an edge $e = (s, s')$ in G_T that takes a robot from s to s' . Suppose there is a terminal or goal state s_T in the graph at which we define a self-loop, $e_T = (s_T, s_T)$, while at other nodes we have outgoing edges (s, s') to different $s' \neq s$. Assume the graph is connected so there is a path from any node to any other node in the graph. Define a stage cost as $\ell(s, s') = 1$ if $s \neq s'$ and $\ell(s_T, s_T) = 0$. We solve the shortest-path problem from s_I to s_T by setting $V_0(s_T) = 0$ and $V_0(s) = 10$ for $s \neq s_T$. Terminate the algorithm at iteration $d + 1$ when we get a stationary value $V_d(s_I) = V_{d+1}(s_I)$ at s_I (other values for other states may not have converged). Answer the following:
 - (a) How many iterations does it take for this algorithm to converge?
 - (b) What does the policy of the resulting $V_d(s)$ guarantee in terms of finding a shortest path from s_I to s_T ?
 - (c) Will the number of iterations change if we initialize the algorithm from a different V_0 ?
3. (Comparison of algorithms) Compare BFS (Breadth FS, the backward search) and the VI algorithm adapted for searches over graphs as in the previous problem. How do they differ in the number of iterations until convergence, computational cost, and generality (whether you can export the same ideas to more general scenarios)?
4. Consider an episodic MDP and a policy π . Suppose we have a sequence of sample returns $G_1, \dots, G_{n-1}$ starting at s and let us set

$$V_n(s) = \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k},$$

where W_k an arbitrary weight, $W_k \geq 0$. Answer the following:

- (a) Reason why $V_n(s)$ is an unbiased estimator of $v_\pi(s)$; that is, show that $\mathbb{E}[V_n(s)] = v_\pi(s)$. What would its variance? *Note:* Assume the variance of a sample return is σ^2 . Recall that the definition of a variance of a random variable is given by $\text{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}(Y))^2]$, and use independence of the sample runs.
- (b) (Extra credit) Is there any advantage in having $W_k \neq \frac{1}{n}$ for the estimation of v_π ? (consider variance reduction.)
- (c) Using the idea of running averages, derive a mechanism to update $V_{n+1}(s)$ from $V_n(s)$, W_n and $C_n = \sum_{k=1}^{n-1} W_k$.
- (d) Adapt the pseudo-code of the first-visit MC prediction, for estimating $v_\pi(s)$ (in the RL book) to i) use every-visit MC, and ii) use this type of updates.